THE ZERO-HUMAN FLEET

HOW WE BUILT A MULTI-AGENT LAB THAT RUNS ITSELF

> From single-agent setup to a coordinated fleet of 10+ autonomous agents. What broke. How we fixed it. What we learned.

> *"The moment you scale past one agent, everything breaks in ways that look like bugs but are actually design assumptions."*
>
> — FIELD NOTES, WEEK 1

10+	5	6	0
AGENTS RUNNING	GATEWAY PROCESSES	PATCHES APPLIED	HUMANS REQUIRED

WHAT WE WERE TRYING TO DO

Run a zero-human lab. Multiple entities, 10+ agents, **zero humans checking in**. Each agent with its own memory, Telegram bot, and cron schedule — all firing simultaneously, all coordinated through shared brain files, none blocking the others. Hermes ships configured for one agent. The moment you scale past that, the defaults actively fight you.

WHAT ACTUALLY BROKE

SEQUENTIAL CRON

CEO heartbeat takes 20 minutes. **Every other agent queued behind it.** BMM missed hour triggers. Siren missed daily batches. Everything stacked behind one job.

SKIP_MEMORY=TRUE

CEO writes to shared brain. Downstream agents read from it. **Channel closes mid-cron.** Lab agent runs on stale data. Silent failure.

MESSAGING BLOCKED

Cron ran. Output was generated. **Nobody saw it.** Default cron disables the messaging toolset entirely. Results never reached Telegram.

HERMES_EXEC_ASK = "1"

Every command paused waiting for human approval at gateway boot. **Cron jobs stopped dead.** Autonomous operation model broken.

TELEGRAM RETRYAFTER

Not a `NetworkError` subclass. Gateway's catch block never caught it. **Messages just stopped.** No error. No retry. Silent death.

KIMI API REJECTION

api.kimi.com only accepts whitelisted User-Agents. **Every call failed silently.** Agents didn't get responses and nobody knew why.

THE ARCHITECTURE

Two types of agents, one codebase.
Gateway agents — full separate OS processes (CEO, Lab, BMM, Siren, BA). Each with its own HERMES_HOME, own Telegram bot, own cron scheduler, own always-on memory system. **Internal cron agents** — in-memory workers (Knowledge Extractor, Team State Engineer, Process Engineer, Epoch Evaluator). Spawned by CEO's cron jobs. Share CEO's memory system, each with isolated profile-scoped working state.

One Hermes install on disk. Six PM2 entries. Six isolated cron contexts. Fleet coordinates through shared brain files.

THE AGENTS

CEO
Root · Opus 4.6 · Orchestrator

LAB GENIUS
MiniMax M2.7 · Full-stack R&D

BMM SUPPORT
MiniMax M2.7 · Customer Email

SIREN
Batch Pipelines · Daily Ops

BUSINESS ANALYST
On-demand · Mention-triggered

KNOWLEDGE EXTRACTOR
Internal · Cron Worker

TEAM STATE ENG.
Internal · Cron Worker

PROCESS ENGINEER
Internal · Cron Worker

EPOCH EVALUATOR
Internal · Cron Worker

WIKI KEEPER
Internal · Cron Worker

THE SIX PATCHES

PATCH-001 · USER-AGENT

Kimis API rejects everything except `claude-code/1.0`. Swap in two locations in `run_agent.py`.

PATCH-002 · CONCURRENT CRON

`ThreadPoolExecutor` runs due jobs in parallel. 20 workers max. Single job skips pool overhead.

PATCH-003 · AUTO-APPROVE

`HERMES_EXEC_ASK="0"` hardcoded. No human in the loop. Cron jobs fire autonomously.

PATCH-004 · MEMORY + MESSAGING

`skip_memory=False` + messaging toolset enabled. Brain writes survive cron. Results deliver to Telegram.

PATCH-005 · FLOOD CONTROL

`RetryAfter` caught explicitly before `NetworkError` handler. Wait the duration. Retry.

PATCH-006 · TIMEOUT BUMP

`HERMES_CRON_TIMEOUT` bumped to 1200s (20 min). CEO heartbeat fits within it.

THE RE-APPLY SCRIPT

Every `hermes update` wipes the codebase and wipes all patches. One script — detects what's already applied, restores what isn't, idempotent by design. `bash ~/company/hermes/patches/reapply-all-patches.sh` Run it after every update. Takes seconds.

WHAT WE'VE BUILT

- **Single-Agent Setup**
  CEO agent running from root ~/.hermes/. One bot, one channel, one heartbeat.
- **Profile Architecture**
  Lab Genius, BMM Support, Siren each given own HERMES_HOME. First isolation layer.
- **Multi-Gateway Fleet**
  Five separate PM2 entries. Five Telegram bots. Five isolated cron schedulers.
- **Internal Workers**
  Knowledge Extractor, Team State Engineer, Process Engineer, Epoch Evaluator spawned as in-memory cron agents.
- **Concurrent Cron + All Patches**
  ThreadPoolExecutor, auto-approve, skip_memory=False, messaging enabled, flood control fixed, timeout bumped.
- **Fleet Operating**
  10+ agents. Zero humans. Coordinated via shared brain files. Field manual published.

LAB EXPERIMENTS

YouTube Shorts Pipeline · Video Context Index · Faceless Channel · Knowledge Graph · Social Scheduler · Cross-Platform Reposter · FB Image Ads · Design Tokens · Soul.md Framework

LAB IMPLEMENTATIONS

Command Center · Deliverables Gallery · Fleet Dashboard · BMM Documentary Noir · Purpose Gateway Editorial · Trading Ultra-Modern · Agent Onboarding Template

KNOWLEDGE BASE

14,500+ Fathom records · 685 transcripts · Strategic reports · Domain intelligence · Brain system (Wiki + Epoch + KE unified)

ENTITIES RUNNING

BMM (Body Magic Method) · CamMitchell Trading · Purpose Gateway

More launching as the fleet coordinates.

THE VISION

A founder who creates at every layer simultaneously.

Every entity, every agent, every process —

running without a single human in the loop.

10+ Agents · Zero Humans · Multi-Entity · Fully Autonomous